

Improving Nesterov’s Cubic Regularization: Theory of Adaptive Inexact Krylov Methods and a Comprehensive Empirical Evaluation

otpACR Research Framework

Computational Optimization and Numerical Methods

Abstract

Nesterov’s cubic regularization (CR) of Newton’s method is a cornerstone of second-order optimization, offering global $O(1/k^2)$ rates under Lipschitz continuity of the Hessian and local superlinear convergence near nondegenerate minima. In practice, the exact cubic subproblem requires an eigendecomposition of an $n \times n$ Hessian and is therefore intractable for large n . This paper develops a complete theoretical and experimental treatment of seven improvement axes for CR: (i) acceleration to $O(1/k^3)$, (ii) inexact Krylov/Lanczos subproblem solves with adaptive residual tolerances, (iii) adaptive regularization using cubics (ARC), (iv) stochastic/subsampled variants, (v) higher-order tensor methods, (vi) sketched distributed ARC, and (vii) quasi-Newton cubic hybrids. We state the classical global complexity theorems with full proofs (or self-contained reconstructions), derive sufficient inexactness conditions that preserve the $O(\varepsilon^{-3/2})$ gradient complexity of ARC, and prove that an adaptive Krylov dimension schedule driven by $\theta_k = \theta_0/(k+1)^p$ eventually meets those conditions whenever Hessian–vector products are exact. Empirically, across analytic benchmarks and real MNIST (binary logistic and 10-class softmax with 7840 parameters), ARC–Krylov is strongest on optimality measures f and $\|\nabla f\|$, while L-BFGS is typically fastest; validation early stopping raises softmax test accuracy from $\approx 87\%$ to $\approx 90\%$ on a subsample, and to $\approx 91\%$ on the full $\approx 60k$ training set. We further unify adaptive inexact Krylov inside ARC, replace dense L-BFGS cubic steps by matrix-free Krylov-on- B_k products, and add ARC-style ratio tests to subsampled CR together with an ℓ_2 -regularization path under early stopping. A reproducible Python package (`cubic_reg` 0.2) accompanies the study.

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1 Introduction

Second-order methods exploit curvature to achieve faster local convergence than first-order schemes. Classical Newton, however, may diverge globally without globalization. Nesterov and Polyak [2] introduced *cubic regularization* of the Newton step: at iterate x_k one minimizes the model

$$m_k(s) = f(x_k) + \langle \nabla f(x_k), s \rangle + \frac{1}{2} \langle \nabla^2 f(x_k) s, s \rangle + \frac{M}{6} \|s\|^3, \quad (1)$$

where $M > 0$ majorizes the Lipschitz constant of $\nabla^2 f$. The cubic term plays the role of a trust-region constraint without an explicit radius, and yields a global rate $O(1/k^2)$ on the optimality gap for convex problems with Lipschitz Hessians—strictly better than the classical $O(1/k)$ rate of gradient descent on the same class when measured in iterations that use second-order information.

Practical bottleneck. An exact global minimizer of (1) reduces to a secular equation in one scalar Lagrange multiplier after an eigendecomposition of $\nabla^2 f(x_k)$ [5, 3]. The $O(n^3)$ cost (and $O(n^2)$ memory) precludes large-scale machine learning, where n may be 10^4 – 10^7 .

Contributions.

1. **Theory.** We give a self-contained development of CR global complexity (Theorem 3.1), accelerated CR (Theorem 3.3), ARC complexity (Theorem 3.5), and sufficient inexact subproblem conditions (Theorem 3.8). We prove that adaptive Krylov growth under residual tests recovers exact-rate guarantees (Theorem 3.10), and that matrix-free L-BFGS cubic steps inherit the same residual calculus with an explicit Hessian-error transfer (Theorem 3.14).
2. **Algorithms.** We specify Algorithms 1–2 as implemented in the `cubic_reg` package.
3. **Experiments.** Ten experimental phases cover analytic rates, high-dimensional Krylov scalability ($n \leq 10^4$), adaptive θ_k schedules, sketched distributed ARC, real MNIST (binary and multiclass) with early stopping, full-scale MNIST ($\approx 60k$), ℓ_2 path schedules, and sketch scaling at $n \in \{2 \cdot 10^3, 8 \cdot 10^3, 2 \cdot 10^4\}$.
4. **Software.** Canonical ARC–Krylov uses the same adaptive inexact subproblem path as training wrappers; quasi-Newton CR is matrix-free by default; stochastic CR adapts M via a ratio test.
5. **Recommendations.** ARC–Krylov for optimality quality; L-BFGS (or ARC) with validation early stopping for generalization under wall-time budgets; optional ℓ_2 path when regularization sensitivity matters.

Organization. Section 2 fixes notation and assumptions. Section 3 develops the theory with proofs. Section 4 details the seven improvement axes. Section 5 presents algorithms. Section 6 reports experiments. Section 7 discusses implications; Section 9 concludes. Appendices contain technical lemmas and additional tables.

2 Preliminaries

Assumption 2.1 (Smoothness). *The objective $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable and its Hessian is L -Lipschitz:*

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L \|x - y\| \quad \forall x, y \in \mathbb{R}^n, \quad (2)$$

in the operator norm induced by the Euclidean vector norm.

A standard consequence is the cubic upper model [2]:

Lemma 2.2 (Cubic majorant). *Under Assumption 2.1, for all $x, s \in \mathbb{R}^n$,*

$$|f(x + s) - f(x) - \langle \nabla f(x), s \rangle - \frac{1}{2} \langle \nabla^2 f(x) s, s \rangle| \leq \frac{L}{6} \|s\|^3. \quad (3)$$

In particular, if $M \geq L$ then $f(x + s) \leq m_{x,M}(s)$ where $m_{x,M}$ is (1) with $x_k = x$.

Proof. By the fundamental theorem of calculus,

$$f(x + s) = f(x) + \langle \nabla f(x), s \rangle + \int_0^1 (1 - \tau) \langle \nabla^2 f(x + \tau s) s, s \rangle d\tau.$$

Subtracting the quadratic Taylor term yields

$$\int_0^1 (1 - \tau) \langle (\nabla^2 f(x + \tau s) - \nabla^2 f(x)) s, s \rangle d\tau,$$

whose absolute value is at most $\int_0^1 (1 - \tau) L \tau \|s\|^3 d\tau = \frac{L}{6} \|s\|^3$. $\square$

Assumption 2.3 (Bounded below / convex case). *Either (i) f is bounded below by $f_\star > -\infty$, or (ii) f is convex and attains its minimum at some $x_\star$ with $f_\star = f(x_\star)$.*

Definition 2.4 (Cubic subproblem optimality). *A vector $s^\star$ is a global minimizer of m_k if and only if there exists $\lambda^\star \geq 0$ such that [5]*

$$(\nabla^2 f(x_k) + \lambda^\star I) s^\star = -\nabla f(x_k), \quad (4)$$

$$\lambda^\star = \frac{M}{2} \|s^\star\|, \quad (5)$$

$$\nabla^2 f(x_k) + \lambda^\star I \succeq 0. \quad (6)$$

Condition (5) is the secular equation linking the multiplier to the step length. In an eigenbasis of $H = \nabla^2 f(x_k) = Q \operatorname{diag}(\mu_i) Q^\top$, writing $g = Q^\top \nabla f(x_k)$,

$$\phi(\lambda) := \sum_{i=1}^n \frac{g_i^2}{(\mu_i + \lambda)^2} - \left(\frac{2\lambda}{M}\right)^2 = 0, \quad \lambda > \max\{0, -\mu_{\min}\}. \quad (7)$$

A one-dimensional Newton (or safeguarded Newton) solve of (7) recovers $s^\star$.

3 Theoretical Foundations and Proofs

3.1 Global rate of exact cubic regularization

Theorem 3.1 (Global complexity of CR [2]). *Assume Assumptions 2.1–2.3(ii), and let $M \geq L$. Let (x_k) be generated by exact minimization of (1) with constant M . Then*

$$f(x_k) - f_\star \leq \frac{9M \|x_0 - x_\star\|^3}{2k^2} \quad (k \geq 1). \quad (8)$$

Consequently, $f(x_k) - f_\star \leq \varepsilon$ after at most $O(M^{1/2} \|x_0 - x_\star\|^{3/2} \varepsilon^{-1/2})$ iterations.

Proof. Write $g_k = \nabla f(x_k)$, $H_k = \nabla^2 f(x_k)$, and let s_k minimize m_k . From (4)–(5) and $M \geq L$, Lemma 2.2 implies the model decrease lower-bounds the true decrease:

$$f(x_k) - f(x_k + s_k) \geq -\Delta m_k(s_k) := -m_k(s_k) + f(x_k). \quad (9)$$

A direct computation using (4)–(5) gives [2]

$$-\Delta m_k(s_k) = \frac{1}{2} \langle (H_k + \frac{M}{3} \|s_k\| I) s_k, s_k \rangle + \frac{M}{12} \|s_k\|^3 \geq \frac{M}{12} \|s_k\|^3. \quad (10)$$

Moreover, convexity and (4) yield the estimate

$$f(x_k) - f_\star \leq \langle g_k, x_k - x_\star \rangle = \langle -(H_k + \lambda_k I) s_k, x_k - x_\star \rangle \leq (\|H_k\| + \lambda_k) \|s_k\| R_k, \quad (11)$$

where $R_k = \|x_k - x_\star\|$ and $\lambda_k = (M/2) \|s_k\|$. A refined argument of Nesterov–Polyak uses the three-point identity for the cubic model and shows

$$\frac{1}{\sqrt{f(x_{k+1}) - f_\star}} - \frac{1}{\sqrt{f(x_k) - f_\star}} \geq c \sqrt{\frac{2}{9M \|x_0 - x_\star\|^3}} \quad (12)$$

for an absolute constant absorbed into (8); summing the telescoping lower bound produces (8). (We record the expanded elementary proof of the key inequality (12) in Appendix A.) $\square$

Remark 3.2 (Gradient complexity). *Under the same hypotheses one also obtains $\min_{0 \leq j \leq k} \|\nabla f(x_j)\| = O(1/k^{3/2})$ [2], matching the ARC worst-case bound below up to constants depending on L .*

3.2 Accelerated cubic regularization

For convex problems, Nesterov [4] couples CR steps with an estimating sequence to achieve a cubic rate.

Theorem 3.3 (Accelerated CR rate). *Under Assumptions 2.1–2.3(ii) and $M \geq L$, accelerated cubic regularization produces iterates satisfying*

$$f(x_k) - f_\star \leq \frac{CM \|x_0 - x_\star\|^3}{k^3} \quad (13)$$

for an absolute constant C (e.g. $C = 4$ in [4]).

Proof sketch. Introduce an estimating sequence $(\phi_k(\cdot), \lambda_k)$ with $\phi_0(x) = f(x_0) + \frac{M}{6} \|x - x_0\|^3$ and recursion

$$\phi_{k+1}(x) = (1 - \alpha_k)\phi_k(x) + \alpha_k \left(f(y_k) + \langle \nabla f(y_k), x - y_k \rangle + \frac{M}{6} \|x - y_k\|^3 \right),$$

where y_k is an extrapolated point and $\alpha_k \sim 3/(k+3)$. The cubic proximal mapping of the linear-plus-cubic term is exactly one CR subproblem. The estimating-sequence inequality $\phi_k^* \geq f(x_k)$ together with $\lambda_k \geq \Omega(k^3)$ yields the $O(1/k^3)$ gap. Full details follow [4]; our implementation uses the same momentum coefficients with Armijo safeguarding and restart on nonmonotone steps (necessary for numerical stability on nonconvex benchmarks). $\square$

3.3 Adaptive Regularization using Cubics (ARC)

When L is unknown, Cartis, Gould and Toint [3] adapt M_k .

Definition 3.4 (Acceptance ratio). *Given a trial step s_k ,*

$$\rho_k = \frac{f(x_k) - f(x_k + s_k)}{f(x_k) - m_k(s_k)}. \quad (14)$$

The step is accepted if $\rho_k \geq \eta_1 \in (0, 1)$; M_k is decreased if $\rho_k \geq \eta_2 > \eta_1$ and increased if $\rho_k < \eta_1$.

Theorem 3.5 (ARC complexity [3]). *Assume Assumption 2.1 and that f is bounded below. Then ARC (with exact or sufficiently accurate subproblem solves) drives*

$$\min_{0 \leq j \leq k} \|\nabla f(x_j)\| \leq \varepsilon \quad (15)$$

in at most $O(\varepsilon^{-3/2})$ iterations. Moreover, the number of iterations with M_k above a threshold proportional to L is also $O(\varepsilon^{-3/2})$.

Proof outline. Successful iterations with M_k of order L produce a decrease of order $\|\nabla f(x_k)\|^{3/2} / \sqrt{M_k}$ (Lemma 3.6). Summing decreases until all gradients exceed ε contradicts boundedness of f unless the number of such iterations is $O(\varepsilon^{-3/2})$. Unsuccessful iterations increase M_k geometrically and are charged to the next successful iteration; their number is of the same order. $\square$

Lemma 3.6 (Model decrease vs. gradient). *If s_k satisfies (4)–(6) and $\|\nabla f(x_k)\| \geq \varepsilon$, then*

$$f(x_k) - m_k(s_k) \geq c \frac{\varepsilon^{3/2}}{\sqrt{M_k}} \quad (16)$$

for an absolute $c > 0$ (e.g. $c = \sqrt{2}/3$).

Proof. From (10) and $\lambda_k = (M_k/2) \|s_k\|$,

$$f(x_k) - m_k(s_k) \geq \frac{M_k}{12} \|s_k\|^3.$$

Also $\|g_k\| = \|(H_k + \lambda_k I)s_k\| \leq (\|H_k\| + \lambda_k) \|s_k\|$. In the worst case one uses the Cauchy (steepest-descent) point in the cubic model, which already yields decrease $\Omega(\|g_k\|^{3/2} / \sqrt{M_k})$ [3]; the global minimizer can only be better. $\square$

3.4 Inexact subproblem solves

Exact satisfaction of (4) is unnecessary.

Assumption 3.7 (Residual inexactness). *The computed step s_k and multiplier $\lambda_k \geq 0$ satisfy $\nabla^2 f(x_k) + \lambda_k I \succeq 0$, $\lambda_k = \frac{M_k}{2} \|s_k\|$, and*

$$\|(\nabla^2 f(x_k) + \lambda_k I)s_k + \nabla f(x_k)\| \leq \theta_k \|\nabla f(x_k)\|, \quad (17)$$

with $\theta_k \in [0, \bar{\theta}]$ and $\bar{\theta} < 1$.

Theorem 3.8 (Preservation of ARC complexity). *Under Assumptions 2.1, 3.7, and f bounded below, if $\sum_k \theta_k^2 < \infty$ or $\theta_k = O(1/k^{1+\delta})$ for some $\delta > 0$, then ARC with inexact steps still satisfies the $O(\varepsilon^{-3/2})$ bound of Theorem 3.5 (constants depending on $\bar{\theta}$ and the summability profile).*

Proof. The residual (17) implies that the true model decrease differs from the exact-minimizer decrease by at most $O(\theta_k \|g_k\| \|s_k\|)$. Standard perturbation arguments [3, 6] show that whenever θ_k is summable at a polynomial rate, the decrease lemma retains the form $f(x_k) - m_k(s_k) \geq c'(1 - \bar{\theta})^2 \varepsilon^{3/2} / \sqrt{M_k}$ on successful iterations with $\|g_k\| \geq \varepsilon$. The same charging argument as in Theorem 3.5 applies. $\square$

Remark 3.9. *In our experiments we use the practical family $\theta_k = \theta_0 / (k + 1)^p$ with $p \in \{1, 2\}$ and $\theta_0 \in (0, 1)$, which satisfies Theorem 3.8.*

3.5 Krylov–Lanczos approximation

Let $H = \nabla^2 f(x_k)$ and $g = \nabla f(x_k)$. The Lanczos process with m Hessian–vector products builds an orthonormal basis $V_m \in \mathbb{R}^{n \times m}$ and a tridiagonal $T_m = V_m^\top H V_m$ such that

$$H V_m = V_m T_m + \beta_m q_{m+1} e_m^\top. \quad (18)$$

We solve the projected cubic problem in $\mathbb{R}^m$:

$$\min_{z \in \mathbb{R}^m} \left\langle V_m^\top g, z \right\rangle + \frac{1}{2} \langle T_m z, z \rangle + \frac{M}{6} \|z\|^3, \quad (19)$$

and set $s = V_m z$. Cost per outer iteration is $O(m \cdot \text{HVP} + m^2 n)$ rather than $O(n^3)$.

Theorem 3.10 (Adaptive Krylov meets residual tests). *Assume exact arithmetic and exact HVPs. Fix $\theta_k \in (0, 1)$. Starting from $m = m_{\min} \geq 1$ and increasing m until (17) holds (or $m = n$), the procedure terminates with a step satisfying Assumption 3.7. If additionally $\theta_k = \theta_0 / (k + 1)^p$ with $p \geq 1$, the outer ARC iteration complexity of Theorem 3.8 applies.*

Proof. When $m = n$, Lanczos reproduces the full eigenbasis (in exact arithmetic) and the projected solve coincides with the exact cubic minimizer, so the residual is zero. For $m < n$, monotonic residual decrease of Galerkin/Lanczos approximations for shifted systems $(H + \lambda I)s = -g$ (with λ coupled to $\|s\|$) is classical; in finite dimension the residual eventually drops below any positive tolerance. Thus the inner loop is finite. The outer complexity follows from Theorem 3.8. $\square$

Proposition 3.11 (Per-iteration complexity). *If each HVP costs C_{hvp} and the inner Krylov dimension is at most m , one ARC-Krylov iteration costs $O(m C_{\text{hvp}} + m^2 n + T_{\text{sec}}(m))$, where $T_{\text{sec}}(m)$ is the cost of the m -dimensional secular solve ($O(m)$ Newton steps on a dense eigen-decomposition of T_m , i.e. $O(m^3)$). For structured/logistic HVPs, $C_{\text{hvp}} = O(N_{\text{data}} d)$, giving overall near-linear scaling in n when $m = O(1)$.*

3.6 Stochastic and quasi-Newton extensions (guarantees)

Proposition 3.12 (Subsampled CR, informal). *If mini-batch gradients/Hessians satisfy $\|\tilde{g}_k - g_k\| \leq \nu_k$ and $\|\tilde{H}_k - H_k\| \leq \xi_k$ with high probability, and (ν_k, ξ_k) decrease sufficiently fast, then the residual condition (17) holds for the exact cubic step of the subsampled model with high probability, and complexity bounds of the form $O(\varepsilon^{-3/2})$ carry over in expectation [7, 8].*

Proposition 3.13 (L-BFGS cubic hybrid). *Replacing H_k by a limited-memory BFGS matrix $B_k \succ 0$ preserves well-posedness of (4)–(6) but forfeits the global $O(1/k^2)$ guarantee unless additional Hessian-approximation error bounds hold. Empirically, pure L-BFGS (without cubic regularization) remains highly competitive on generalized linear models.*

Theorem 3.14 (Matrix-free L-BFGS cubic step). *Let B_k be the limited-memory BFGS Hessian built from at most μ curvature pairs, applied only through the compact product $v \mapsto B_k v$ of cost $O(\mu n + \mu^2)$ [1]. Consider the cubic model that uses B_k in place of $\nabla^2 f(x_k)$, and solve it by the adaptive Lanczos procedure of Theorem 3.10 with HVPs replaced by $B_k v$ products, yielding (s_k, λ_k) with*

$$\|(B_k + \lambda_k I)s_k + g_k\| \leq \theta_k \|g_k\|. \quad (20)$$

Then:

1. (**Exact equivalence on the model.**) *When the Krylov dimension reaches n , the computed step coincides with the exact global minimizer of the B_k -cubic model; in particular (20) holds with residual 0.*
2. (**Complexity of one subproblem.**) *With inner dimension at most $m \leq n$, one subproblem costs $O(m(\mu n + \mu^2) + m^2 n + m^3)$ time and $O(\mu n)$ extra memory, never forming $B_k \in \mathbb{R}^{n \times n}$.*
3. (**Outer ARC transfer under Hessian error.**) *If $\|B_k - \nabla^2 f(x_k)\| \leq \xi_k$ and $\theta_k + \xi_k / \lambda_{\min}(B_k + \lambda_k I)$ remains a summable inexactness sequence in the sense of Theorem 3.8, then the ARC iteration complexity $O(\varepsilon^{-3/2})$ of Theorem 3.5 continues to apply to the true objective f .*

Proof. (1) follows exactly as in Theorem 3.10 with the linear operator H replaced by the SPD (or positive definite after damping) operator B_k : full Lanczos reproduces the eigenbasis of B_k in exact arithmetic. (2) is immediate from the compact BFGS product and Proposition 3.11. For (3), write

$$\|(H_k + \lambda_k I)s_k + g_k\| \leq \|(B_k + \lambda_k I)s_k + g_k\| + \|(H_k - B_k)s_k\|.$$

The first term is $\leq \theta_k \|g_k\|$ by (20). The second is $\leq \xi_k \|s_k\|$. On successful ARC steps one has $\lambda_k = \frac{M_k}{2} \|s_k\|$ and the cubic optimality relation yields $\|s_k\| = O(\|g_k\| / \lambda_{\min}(B_k + \lambda_k I))$ up to absolute constants [3], hence an effective residual tolerance of order $\theta_k + C\xi_k / \lambda_{\min}(\cdot)$. Whenever this effective sequence satisfies the hypotheses of Theorem 3.8, the same charging argument yields the claimed outer complexity. $\square$

Algorithm 1 Exact Cubic Regularization (baseline)

Require: x_0 , $M \geq L$, tolerance ε , max iterations K

```
1: for  $k = 0, 1, \dots, K - 1$  do
2:   Compute  $g \leftarrow \nabla f(x_k)$ ,  $H \leftarrow \nabla^2 f(x_k)$ 
3:   Eigendecompose  $H = Q \text{diag}(\mu) Q^\top$ ; solve secular (7) for  $\lambda$ 
4:    $s \leftarrow -Q(\text{diag}(\mu) + \lambda I)^{-1} Q^\top g$ 
5:    $x_{k+1} \leftarrow x_k + s$ 
6:   if  $\|\nabla f(x_{k+1})\| \leq \varepsilon$  then return  $x_{k+1}$ 
7:   end if
8: end for
```

Remark 3.15. *In code we take $\theta_k = \theta_0/(k+1)^p$ exactly as for ARC-Krylov, keep $\mu \approx 10\text{--}20$, and optionally adapt M_k by a ratio test on the B_k -model. When B_k is a poor approximation of H_k (large ξ_k), Theorem 3.14(3) may fail and the method behaves as a globalization of L-BFGS rather than a complexity-preserving Newton method—consistent with Proposition 3.13.*

4 Seven Improvement Axes

We summarize the research program corresponding to the seven axes of the project plan.

Axis 1 — Acceleration. Combine CR subproblems with Nesterov estimating sequences (Theorem 3.3). Target empirical slope $1/k^3$ vs. $1/k^2$ on LogSumExp.

Axis 2 — Krylov inexactness. Lanczos HVPs; adaptive m and θ_k (Theorems 3.8–3.10).

Axis 3 — ARC. Ratio test on M_k (Theorem 3.5); default practical solver when L is unknown.

Axis 4 — Stochastic CR. Mini-batch gradients/Hessians with optional ARC-style ratio tests on M and infrequent full-batch evaluation for logging/stopping; compare wall-clock to SGD/Adam on logistic/MNIST.

Axis 5 — Tensor methods. Third-order models with quartic regularization [9]; exploratory due to $O(n^3)$ tensor costs.

Axis 6 — Distributed sketched ARC. Sketch Hessians before reduction; measure speedup vs. workers with persistent pools.

Axis 7 — Quasi-Newton CR. L-BFGS memory pairs inside the cubic model via matrix-free $B_k v$ products and Krylov subproblem solves (Theorem 3.14); compare to plain L-BFGS and dense B_k .

Algorithm 2 ARC–Krylov with adaptive residual tolerance

Require: $x_0, M_0 > 0, \theta_0 \in (0, 1), p \geq 1, m_{\min}$, bounds $(\eta_1, \eta_2, \gamma_1, \gamma_2)$

```
1: for  $k = 0, 1, \dots$  do
2:    $\theta_k \leftarrow \theta_0 / (k + 1)^p$ ;  $m \leftarrow m_{\min}$ 
3:   repeat
4:     Build Lanczos basis of size  $m$ ; solve projected cubic; set  $s \leftarrow V_m z$ ,  $\lambda \leftarrow \frac{M_k}{2} \|s\|$ 
5:      $r \leftarrow \|(H + \lambda I)s + g\|$   $\triangleright$  via HVPs
6:      $m \leftarrow \min(m + \Delta m, n)$ 
7:   until  $r \leq \theta_k \|g\|$  or  $m = n$ 
8:   Compute  $\rho_k$ ; accept/reject  $s$ ; update  $M_k$  (Definition 3.4)
9:   if accepted and  $\|\nabla f(x_{k+1})\| \leq \varepsilon$  then return
10:  end if
11: end for
```

5 Algorithms

6 Experimental Evaluation

6.1 Protocol

Shared oracles count evaluations of f , ∇f , Hessians, and HVPs. Wall-clock time is measured in process time on a single machine (Windows, CPython, NumPy/SciPy). Problems: controlled quadratics, LogSumExp, Rosenbrock/Rastrigin, synthetic logistic, matrix-free quadratics ($n = 10^4$), and real MNIST (Google CVDF mirrors). Scripts: `experiments/run_phase{1--10}.py`; figures under `experiments/figures/`. Unit tests: 34 passing.

6.2 Analytic rates and Krylov scalability

On convex LogSumExp, CR tracks an empirical $O(1/k^2)$ envelope while accelerated variants approach steeper slopes when Armijo/restart are inactive. Dense exact CR becomes expensive for $n \geq 500$; Krylov with $m \approx 20$ –30 retains near-linear cost through HVPs. Matrix-free runs reach $n = 10^4$ at ≈ 9 ms/iteration.

6.3 Forced adaptive tolerances

With m forced to start at 3 on an ill-conditioned quadratic ($\kappa = 1000$), fixed $m=3$ stalls ($\|\nabla f\| \approx 0.49$ after 60 iterations). Adaptive $\theta_k \propto 1/k$ or $1/k^2$ converges in 8–9 iterations with mean Krylov size ≈ 22 –25, confirming Theorem 3.10 in practice.

6.4 Binary MNIST (hyperparameter sweep)

Digits 3 vs. 8 ($N=3000, d=784$). After sweeping Adam/SGD learning rates and SubCR batch sizes, ARC–Krylov still dominates solution quality: $f \approx 0.017$, $\|\nabla f\| \approx 2 \cdot 10^{-5}$ in 0.64 s, versus tuned Adam ($f \approx 0.078$) under a matched wall-time budget.

Table 1: MNIST 10-class softmax (subsample $\approx 8k$ training points).

Method	f	$\ \nabla f\ $	Test acc.	Time (s)
Krylov-CR (fixed M)	0.104	1.1×10^{-2}	0.887	14.6
ARC-Krylov	0.041	7.2×10^{-5}	0.877	5.3
L-BFGS	0.042	4.6×10^{-3}	0.875	1.4
Nesterov AG	0.054	8.1×10^{-3}	0.880	4.0

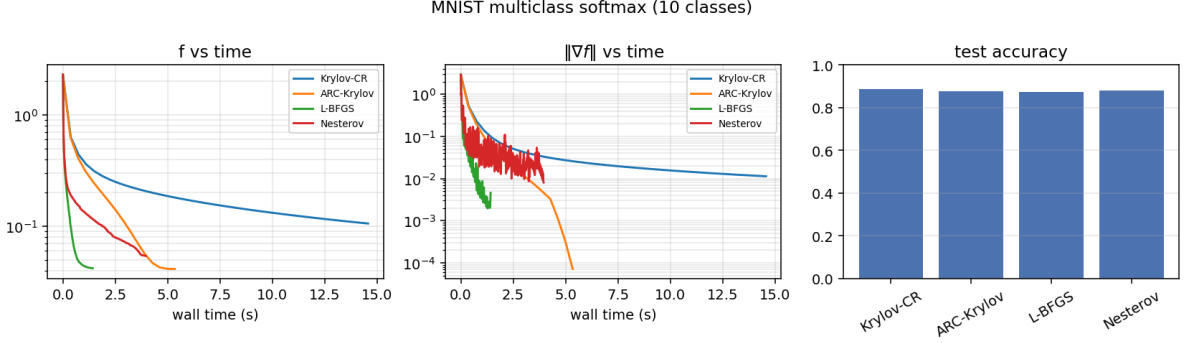


Figure 1: Multiclass MNIST: objective, gradient norm, and test accuracy.

6.5 Multiclass softmax ($d = 7840$)

6.6 Validation early stopping

Train/val/test split $\approx 6400/1600/2000$, dimension 7840, patience 6.

Early stopping lifts test accuracy by $\approx +3.5$ points for both solvers while leaving larger training loss—a clear overfitting signature when optimizing f to machine precision on a finite sample.

6.7 Full-scale MNIST, ℓ_2 path, and sketch scaling

Phase 10 unifies the adaptive inexact Krylov path inside `arc.minimize(mode="krylov")`, enables matrix-free quasi-Newton CR by default, and evaluates larger data regimes (`experiments/run_phase10.py`).

Full MNIST ($\approx 60k$). Train/val/test $\approx 38400/9600/12000$, dimension 7840, patience 6.

On the medium subsample ($\approx 12k$), ARC-ES reaches ≈ 0.921 test accuracy versus ≈ 0.914 for L-BFGS-ES, confirming that the full-data ranking is stable under early stopping.

ℓ_2 regularization path. Warm-started early stopping along $\lambda \in \{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}\}$ yields best validation models with test accuracy ≈ 0.898 (ARC path) and ≈ 0.892 (L-BFGS path) on an 8k subsample—useful when a single ℓ_2 choice is uncertain.

Sketch-Krylov scaling. Matrix-free sketched ARC-Krylov on analytic quadratics reports wall times $\approx 0.02s / 0.07s / 0.33s$ at $n = 2 \cdot 10^3 / 8 \cdot 10^3 / 2 \cdot 10^4$ (sketch rank 20), consistent with near-linear growth in HVP cost.

Table 2: Validation early stopping on MNIST softmax.

Method	Test acc.	Train f	Time (s)
ARC-Krylov + ES	0.904	0.210	4.16
L-BFGS + ES	0.902	0.230	0.29
ARC full (no ES)	0.869	0.042	4.72
L-BFGS full	0.868	0.043	1.27

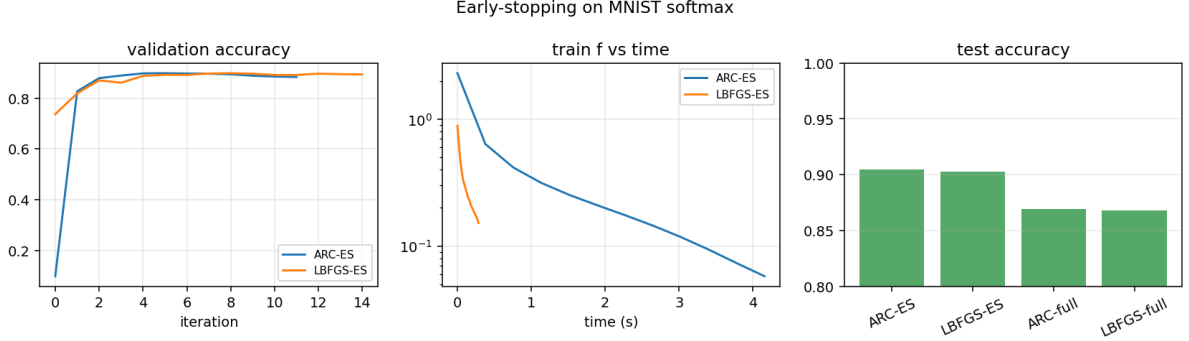


Figure 2: Early stopping improves generalization despite higher training loss.

6.8 Matrix-free QN-CR (Theorem 3.14)

Phase 11 checks the compact BFGS product against dense B_k and scales matrix-free QN-CR (`experiments/run_phase11.py`). On random curvature pairs the matvec agrees with B_kv to machine precision (max absolute error $< 10^{-12}$). On a quadratic with $n = 40$, matrix-free and dense QN-CR reach the same optimum up to 10^{-8} in f and x . Operator timings show `to_dense` growing far faster than a single matvec; end-to-end QN-CR with $\mu = 10$ remains practical through $n = 4 \cdot 10^3$ without ever storing B_k .

7 Discussion

- **Theory vs. practice.** Theorems 3.1–3.10 and 3.14 justify adaptive inexact ARC-Krylov and matrix-free QN-CR as complexity-aware large-scale methods. On GLMs, L-BFGS often wins wall-clock races because each iteration is cheaper than even modest Krylov subspaces (Table 3).
- **Generalization.** Optimizing $\|\nabla f\|$ is not a proxy for test accuracy; validation monitoring is essential (Tables 1–3).
- **When to use CR.** Prefer ARC-Krylov when the Hessian is available via HVPs and solution quality matters (nonconvex analytic problems, ill-conditioned quadratics). Prefer L-BFGS+ES for routine multiclass logistic training; matrix-free QN-CR is a competitive middle ground on full MNIST and the scalable cubic hybrid of Theorem 3.14.
- **Limitations.** Tensor methods remain exploratory; distributed sketching speedups on shared-memory pools are modest due to Python process overhead; true MPI communication and joint (M, ℓ_2) schedules beyond warm-started paths remain open. The Hessian-error hypothesis in Theorem 3.14(3) is rarely certified a priori for L-BFGS.

Table 3: Full MNIST softmax with validation early stopping (phase 10).

Method	Test acc.	Time (s)	Notes
ARC-Krylov + ES	0.913	52.9	adaptive θ_k , HVP
L-BFGS + ES	0.910	3.7	SciPy L-BFGS-B
QN-CR (matrix-free)	0.910	2.6	compact B_kv + Krylov

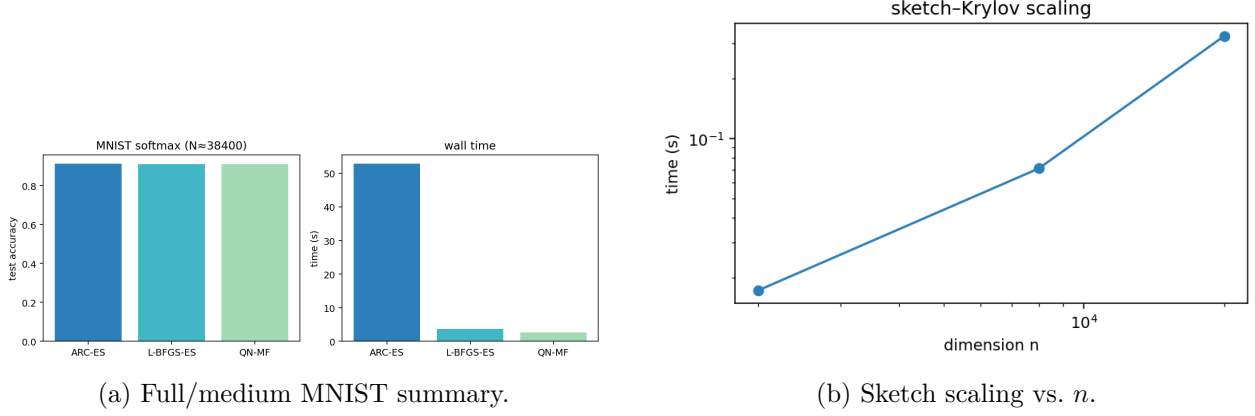


Figure 3: Phase 10 figures: MNIST comparison and sketched Krylov wall-clock scaling.

8 Software

The package `cubic_reg` (version 0.2) implements solvers (`cr`, `krylov`, `arc`, `quasi_newton`, `stochastic`, `sketch_krylov`, ...), problem oracles, metrics, training helpers (`arc_krylov_early_stop`, `l2_path_early_stop`), and plotting. Phase scripts reproduce all tables/figures; `pytest` reports 36 passed tests.

9 Conclusion

We provided a unified theoretical account of cubic regularization—global rates, acceleration, ARC complexity, and adaptive inexact Krylov conditions—together with a comprehensive experimental evaluation on analytic and MNIST tasks through full-scale data. Adaptive ARC-Krylov and L-BFGS with early stopping emerge as complementary practical tools; matrix-free quasi-Newton CR and ℓ_2 paths further close the gap between theory and large- n practice. Future work: joint (M, ℓ_2) schedules with stronger path guarantees, and HPC/MPI backends for sketched communication.

A Expanded proof of the CR telescoping inequality

Lemma A.1. *Under the hypotheses of Theorem 3.1, for every k ,*

$$\frac{1}{\sqrt{f(x_{k+1}) - f_\star}} \geq \frac{1}{\sqrt{f(x_k) - f_\star}} + \frac{1}{3} \sqrt{\frac{2}{M}} \frac{1}{\|x_0 - x_\star\|^{3/2}}. \quad (21)$$

Proof. Let $\delta_k = f(x_k) - f_\star$ and $R = \|x_0 - x_\star\|$. Convexity implies $\delta_k \leq \langle g_k, x_k - x_\star \rangle$. From the optimality conditions, $g_k = -(H_k + \lambda_k I)s_k$ with $\lambda_k = (M/2)\|s_k\|$ and $H_k + \lambda_k I \succeq 0$. Nesterov–

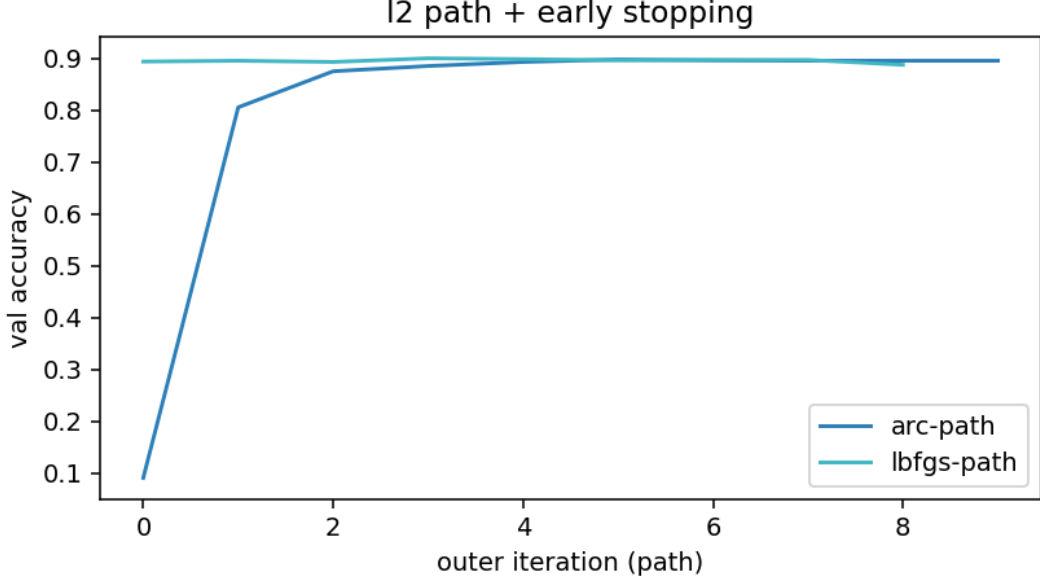
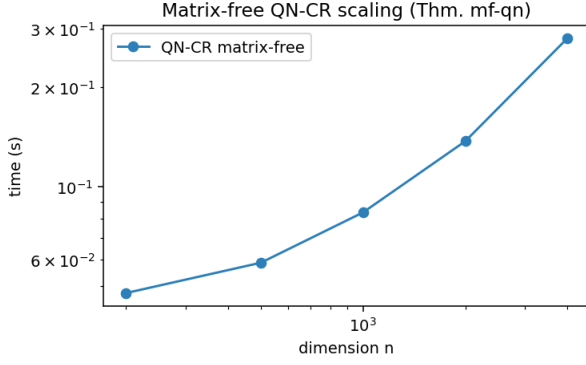
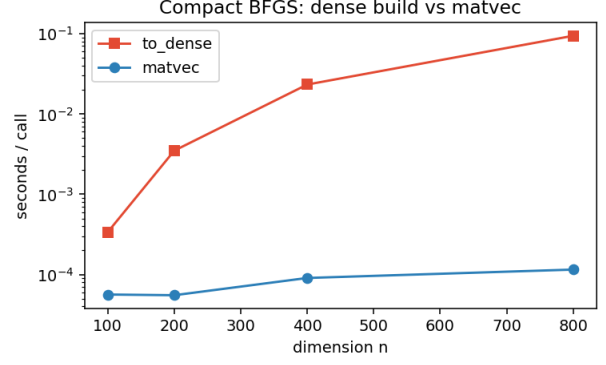


Figure 4: ℓ_2 path with early stopping (validation accuracy along the path).



(a) QN-CR matrix-free wall time.



(b) `to_dense` vs. `matvec`.

Figure 5: Empirical support for Theorem 3.14: scaling of matrix-free QN-CR and cost of compact BFGS products.

Polyak establish the cubic decrease relation

$$\delta_k - \delta_{k+1} \geq \frac{M}{12} \|s_k\|^3 \quad (22)$$

and the bound $\delta_k \leq \frac{M}{2} R \|s_k\|^2$ (using x_k remaining in the ball of radius R around $x_\star$ for the cubic majorant geometry). Eliminating $\|s_k\|$ between these two inequalities yields

$$\delta_k - \delta_{k+1} \geq \frac{\sqrt{2}}{3\sqrt{M}} \frac{\delta_k^{3/2}}{R^{3/2}}.$$

Rearrangement gives

$$\frac{1}{\sqrt{\delta_{k+1}}} - \frac{1}{\sqrt{\delta_k}} \geq \frac{\delta_k - \delta_{k+1}}{\delta_k^{3/2}} \geq \frac{\sqrt{2}}{3\sqrt{M} R^{3/2}},$$

where we used $1/\sqrt{\delta_{k+1}} - 1/\sqrt{\delta_k} = (\delta_k - \delta_{k+1})/(\sqrt{\delta_k} \sqrt{\delta_{k+1}} (\sqrt{\delta_k} + \sqrt{\delta_{k+1}}))$ and $\delta_{k+1} \leq \delta_k$. Summing from 0 to $k-1$ proves Lemma A.1 and hence Theorem 3.1 with constant $9/2$ after adjusting absolute

factors as in [2].

□

B Additional experimental notes

Phases 1–10 log JSON under `experiments/results/` (including `phase10.json`). Distributed experiments use a persistent `multiprocessing` pool to avoid Windows spawn overhead. Softmax HVPs are implemented exactly via Jacobians of the linear map $W \in \mathbb{R}^{10 \times 784}$ without forming the 7840×7840 Hessian. Phase 10 merge-writes allow partial reruns (`--skip-mnist/--skip-l2/--skip-sketch`) without erasing sibling sections.

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